

Customer Segmentation Report

NOVAmarket

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Executive Summary

Using customer segmentation and market basket analysis, we successfully divided NOVAmarket's customer base of approximately 30 thousand customers into **five distinct behavioral segments**. These segments reveal substantial differences in loyalty, spending habits, promotion sensitivity, and product preferences. By combining clustering techniques with association-rule mining, we designed tailored marketing campaigns aimed at increasing customer retention, boosting basket value, and improving promotional effectiveness.

The combination of segmentation and basket analysis provides NOVAmarket with a scalable framework for personalized marketing. Implementing the proposed campaigns and operational recommendations can improve customer retention, increase average transaction values, and optimize promotional spending across the customer lifecycle.

Exploratory Data Analysis & Pre-Processing

When our team picked the customers dataset, the data wasn't yet ready for applying clustering algorithms, so we had to focus on the exploration and preprocessing of our different variables in order to ensure data quality.

The majority of the features presented between 0% and 3% missing values, allowing us to impute those values with the median.

The Year of the First Transaction was also a feature that could contain some level of inconsistency, as the majority of the distribution is between the years 2000 and 2020, having then a huge unexplained spike in 2021, it could resemble a post-COVID recovery. And then the records ever since 2022 until 2025 are minimal compared to

the previous 2 decades. This feature also includes records until 2029, which is in the future.

This could all be explained by the COVID shocks and even the majority of stores closing or weaker marketing campaigns. We decided to leave those records (from 2000 until 2026), as the recorded year might be miswritten, but the actual records could be true. We just changed the date to correspond to the current year of 2026.

Then we could also engineer some new features like the proportion of fresh food bought on the total spendings (as this feature can aggregate 3 to 4 other features), a feature that enhances higher spenders driven by promotions (important to locate potential targets), or another feature that locates loyal customers. These metrics weren't directly shown in the original dataset, so with these engineered features, we can now locate new profiles of customers.

Customer Segmentation & Clustering

In order to decide the number of customer segments (clusters) that it would be reasonable to build, we built and cross-verified both the Elbow Method and the Silhouette Score on top of a series of K-Means algorithm implementations. From those, we took that there are 5 logical clusters.

K-means was tested with and without PCA applied in our data. Even though, with PCA, we would get more clusters (more specifically, 8). Although this could benefit the customers (by having more campaigns), we decided to stick with the K-Means of 5 clusters in order to not lose interpretability.

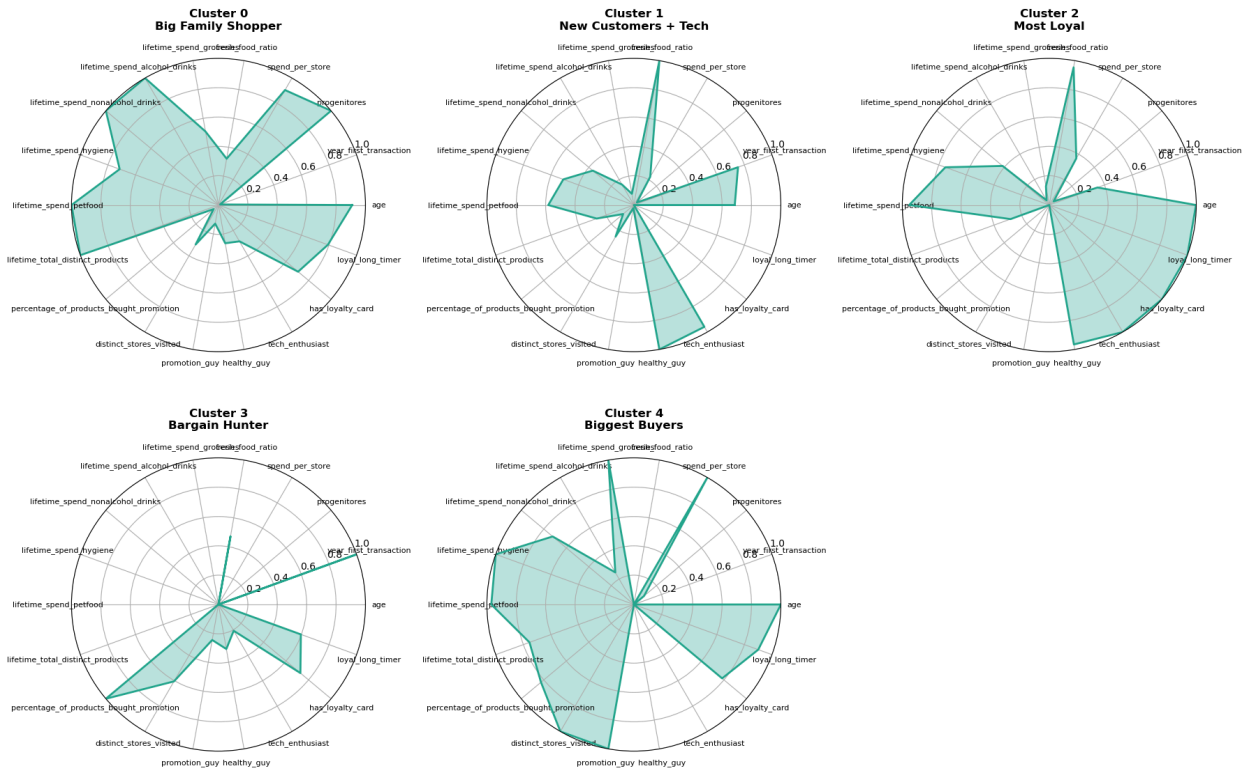
Other algorithms were tried (with and without PCA), and the conclusions of the modeling told us that for most algorithms, the right number of clusters would be between 2 and 4, depending on the algorithm applied.

Therefore, the best solution for this segmentation was K-Means with 5 clusters. The clusters found were:

- **Big Family** | Easily identifiable by the high number of teens and children at home, they tend to have more drinks and pet food bought and they also follow a trend of being somewhat loyal
- **New Customers + Tech** | Those are the ones that are relatively new customers with a tendency to buy healthy and tech items and even some pet food. This might suggest those are younger individuals that follow lifestyle trends
- **Most Loyal** | These are the customers that are the most loyal, they spend money in fresh food, pet food and even hygiene
- **Bargain Hunter** | Definitely the ones that are willing to go to other shops in order to get the best promotions

- **Biggest Buyers** | Here we have probably people that run local stores or restaurant owners, they are also loyal in the sense that they buy a lot of NOVAmarket items

Cluster Profiles - Radar Chart



Targeted Promotion

Big Families

- **Profile:** Large households characterized by high multi-person volumes. They prioritize staple breakfast routines, extensive beverage selections (both alcoholic and non-alcoholic), and consistent personal hygiene spending.
- **Insights:** This segment yields a highly structured, healthy breakfast network, showcasing an intuitive nutritional affinity:
 - Eggs + Green Tea → Cereals (Lift: 1.37)
 - Cereals + Eggs → Green Tea (Lift: 1.36)
 - Honey → Cereals + Oatmeal (Lift: 1.32)
- **Proposed Campaigns**
 - **"The Complete Family Breakfast Table":** Purchase cereals, green tea, and eggs together to instantly unlock a 30% discount on raw honey and premium oatmeal.

- **"Weekend Host / Party-Ready Bundle"**: Buy any 3 high-margin alcoholic beverages and claim a 20% discount across non-alcoholic mixers and sodas.
- **"Pantry Staple Hygiene Pack"**: Spend a minimum of €30 within the personal hygiene category and receive the lowest-priced item completely free.

Fresh Tech Newcomers

- **Profile**: Newer accounts showing immense financial potential via modern tech adoption and a strong fresh food ratio, yet lacking membership enrollment. Their base baskets are heavily dominated by high-frequency routine grooming products.
- **Insights**: A tight, high-confidence grooming loop dominates their basket structures, indicating programmatic bathroom routines:
 - Shower Gel → Shampoo (Lift: 3.44)
 - Deodorant → Shower Gel (Lift: 3.44)
 - Toothbrush → Shower Gel (Lift: 3.33)
 - Shampoo → Toothbrush (Lift: 3.19)
- **Campaigns**:
 - **"The Daily Hygiene Kit"**: Bundle shampoo, shower gel, and deodorant into a single purchase to receive a complimentary toothbrush.
 - **"The Fresh Tech Ecosystem"**: Purchase €50 or more in high-margin fresh food items and unlock a 10% discount voucher valid for any electronics or videogames purchase.
 - **"Welcome to the Club"**: Register for a store loyalty card today to activate an immediate 15% discount on your tech or hygiene basket (eligible reward category rotates weekly).

Stable Loyalists

- **Profile**: The ideal customer standard. This core segment features 100% loyalty card adoption, sustained historic longevity, and broad cross-department spending across Fresh Food, Hygiene, and Consumer Technology.
- **Insights**: Basket analyses expose clear cross-department links connecting consumer tech accessories with impulse energy stimulants:
 - Energy Drink → AirPods (Lift: 2.61)
 - Bluetooth Headphones → AirPods (Lift: 2.49)
- **Campaigns**:
 - **"The Commuter Tech Bundle"**: Purchase a pair of AirPods or premium Bluetooth headphones and receive a free chilled energy drink pack at checkout.
 - **"Loyal Member Tiered Rewards"**: Swipe your verified Loyalty Card to activate an exclusive 10% discount across changing focal categories

(Fresh Food / Personal Care / Electronics) on a weekly promotional cycle.

Bargain Hunters

- **Profile:** Highly price-sensitive consumers who systematically optimize their spending habits toward items that feature active discounts. They heavily index into fresh produce.
- **Insights:** Exceptional clustering performance across raw, fresh vegan ingredients paired with tech accessories:
 - Asparagus + Carrots → Salad (Lift: 3.56)
 - Salad → Asparagus + Avocado (Lift: 3.52)
 - Asparagus + Carrots → Avocado (Lift: 3.45)
 - Spinach → Asparagus + Carrots (Lift: 3.28)
 - Bluetooth Headphones → Energy Drink (Lift: 3.27)
- **Campaigns:**
 - **"The Organic Healthy Plate Deal":** Purchase fresh asparagus, packaged salad, and spinach together to receive carrots and avocados at half price.
 - **"Double Promotion Wednesdays":** Visit store locations mid-week to claim an extra stacked 10% discount on all items already carrying baseline promotions.

Biggest Buyers

- **Profile:** Those consistent and high-spending local shop owners or restaurant owners.
- **Insights:** Market basket analysis reveals that those are consistent buyers that spend huge amounts of money.
 - Asparagus → AirPods (Lift: 1.07)
- **Campaigns:**
 - **"The Elite Platinum Tier Recognition" (Basket Threshold Reward) ***
Mechanics: Exceed a minimum total transaction value of €150 on any visit to unlock an unconditional, flat 15% discount across your entire checkout receipt.

Conclusion

This analysis segmented NOVAmarket's 30,000 customers into **five distinct groups**, each with unique purchasing behaviors, loyalty patterns, and responses to promotions. By combining customer segmentation with market basket analysis, we identified targeted marketing opportunities that can improve customer retention, increase basket value, and optimize promotional spending.

The most immediate opportunity is to increase loyalty program adoption among **Fresh Tech Newcomers** while retaining **Stable Loyalists** through exclusive rewards. **Bargain Hunters** should receive targeted promotions, whereas **Big Families** and **Biggest Buyers** can be encouraged to increase spending through bundles and premium offers.

Overall, NOVAmarket should shift from broad, one-size-fits-all promotions towards **personalized, segment-driven campaigns**, allowing marketing resources to be used more effectively and generating stronger long-term customer value.

Repository: <https://github.com/gugaffranco/bestrepoever>